Lesson 10: Play It Again

TIC-80

VIDEO GAME CREATION

This week: when the round ends, the game actually stops - and pressing R brings it back to a genuinely fresh start.

Open your own [YourName]Game project from Lesson 9 - don't start a new one.

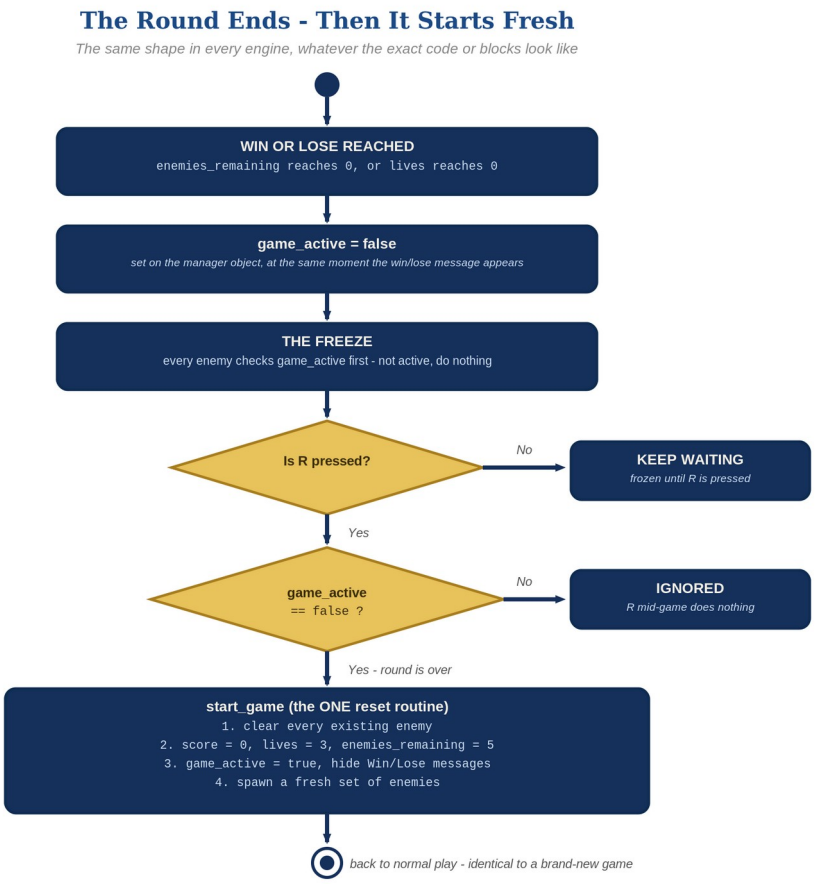
Tip: no new assets this week - this is a logic lesson, and it reuses your very first setup code from Lesson 5.

Setup

1. Launch TIC-80 and load your cart from Lesson 9.
2. Press Esc to reach the code editor.

Level 1 - Minimal (MVP)

When the round ends (win or lose), the game freezes - score and lives stop changing, and further enemy hits/escapes do nothing. Pressing R resets everything and starts a new round.



When the round ends, everything freezes - and R, pressed only while the round is over, runs the one reset routine. The same shape in every engine.

3. Add game_active=true near the top of your code, with your other game state (score, lives, enemies_remaining).
4. Find your existing one-time setup code (populating the enemies table, setting lives, score, enemies_remaining). Move it into a new function start_game() (code below), called once at the top level in place of the old inline setup.
5. In the enemy-drawing loop, wrap the movement and escape code in "if game_active then ... end" - this freezes movement once the round ends.
6. In the bullets loop's hit branch and the enemy loop's escape branch, when enemies_remaining or lives reaches zero, also set game_active=false, and change the print() text to include the restart hint.
7. At the bottom of TIC(), add a check using keyp(18) (TIC-80's key code for R) that calls start_game() if game_active is false (code below).
8. Press Esc, type run, press Enter, then play to a win or loss, confirm movement freezes, then press R and confirm everything resets and play resumes.

Highlighted lines are new this week; everything else is code you already have.

```
-- player position x and y are set above this, exactly as in earlier
```

```
lessons
game_active=true

function start_game()
 enemies={}
 for enemy_count=0,4 do
  table.insert(enemies,{ex=20+enemy_count*20,ey=10,alive=true})
 end
 bullets={}
 score=0
 lives=3
 enemies_remaining=5
 game_active=true
end

start_game()

function TIC()
 cls(0)
 if btn(3) then x=x+1 end
 if btnp(4) then
  table.insert(bullets,{bx=x,by=y})
 end
 for bullet_index=#bullets,1,-1 do
  local b=bullets[bullet_index]
  b.by=b.by-2
  local hit=false
  for enemy_index=1,#enemies do
   local e=enemies[enemy_index]
   if e.alive and b.bx<e.ex+8 and b.bx+8>e.ex and b.by<e.ey+8 and
b.by+8>e.ey then
    e.alive=false
    enemies_remaining=enemies_remaining-1
    score=score+10
    if enemies_remaining<=0 then game_active=false end
    hit=true
   end
  end
  if hit then
   table.remove(bullets,bullet_index)
  elseif b.by<0 then
   table.remove(bullets,bullet_index)
  else
   spr(2,b.bx,b.by,0)
  end
 end
 for enemy_index=1,#enemies do
  local enemy=enemies[enemy_index]
  if enemy.alive then
   if game_active then
    enemy.ey=enemy.ey+0.3
    if enemy.ey>136 then
     enemy.alive=false
     lives=lives-1
    if lives<=0 then game_active=false end
    end
```

```
    end
     spr(3,enemy.ex,enemy.ey,0)
  end
 end
 spr(1,x,y,0)
 print("Score: "..score,5,5,12)
 print("Lives: "..lives,5,15,12)
 if enemies_remaining<=0 then
 print("You Win! Press R to Restart",50,60,12)
 end
 if lives<=0 then
 print("Game Over - Press R to Restart",40,60,8)
 end
if keyp(18) and not game_active then
 start_game()
end
end
```

Level 2 - Complete (Alpha/Beta)

A restart is a genuinely clean slate - no leftover enemies, no stale score/lives, and it works correctly no matter how many times in a row you play.

- Play to a win, confirm movement freezes, then press R and confirm score/lives/enemies_remaining are exactly back to 0/3/5 with no leftover live entries in the enemies table.
- Repeat for a loss - same clean reset.
- Press R while the game is still active and confirm NOTHING happens - the "and not game_active" check is what prevents an accidental reset.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Visual: briefly flash the screen (cls() with a different color for one frame) the moment game_active flips to false.
- Structure: notice start_game() is now the single place that sets up a fresh game - called once when the cart first loads, and again from inside TIC() on every restart. No setup code is duplicated anywhere.

If Something's Wrong

- Restart does nothing when you press R: confirm you're using keyp(18) (edge-triggered, fires once per press) rather than key(18) (fires every frame the key is held) - using key() here would call start_game() repeatedly while R is held.
- Restart leaves old enemies on screen: confirm start_game() actually replaces the enemies table with a fresh one (enemies={}) rather than trying to modify the old one in place.
- Enemies still move after game-over: confirm the "if game_active then ... end" wrapper actually surrounds the movement/escape code inside the enemy-drawing loop, not just the drawing (spr()) call.

Want More?

Student Resource List > TIC-80 section, or tic80.com